Number 10 — Rational and irrational numbers · Surds

No calculator. Marks in brackets. ★ = harder question. Leave answers as exact surds in their simplest form.

1 Simplify [4]

a $\sqrt{80} - \sqrt{20}$ **b** $\sqrt{\dfrac{81}{49}}$

2 Simplify [4]

a $\sqrt{18} + 3\sqrt{8}$ **b** $2\sqrt{75} - \sqrt{27}$

3 Expand and simplify [4]

a $(3 - \sqrt{2})^2$ **b** $(2 + \sqrt{5})(2 - \sqrt{5})$

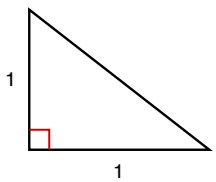
4 Rationalise the denominator and simplify [4]

a $\dfrac{2 + \sqrt{3}}{\sqrt{3}}$ **b** $\dfrac{5}{3 - \sqrt{7}}$

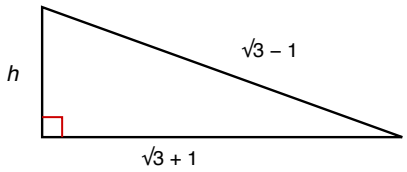
5 Work out, giving each answer in the form $\dfrac{a\sqrt{2}}{b}$: [4]

a $\dfrac{5}{\sqrt{2}} + \dfrac{3}{\sqrt{8}}$ **b** $\dfrac{10}{\sqrt{50}} + \dfrac{3}{\sqrt{2}}$

6 The diagram shows an isosceles right-angled triangle with two sides of length 1. **a** Use it to write $\sin 45°$ as a surd in its simplest form. **b** A boat sails 24 km on a bearing of 045°. How far east has it travelled? Give an exact answer. [2]



7 A right-angled triangle has shorter sides $\sqrt{3}+1$ and $\sqrt{3}-1$. Find the exact length h of the hypotenuse. [3]



Total: 25 marks

Check: $\sqrt{ab}=\sqrt{a}\sqrt{b}$ — pull out the largest square factor · add/subtract only like surds · $(a+\sqrt{b})(a-\sqrt{b})=a^2-b$ · rationalise $\frac{1}{\sqrt{b}}$ with $\sqrt{b}$, and $\frac{1}{a\pm\sqrt{b}}$ with $a\mp\sqrt{b}$ · a surd answer is exact; a decimal is not.